WORKED SOLUTION

STEP III 1999, Question 6

Pure • Parametric Curves and Area

STEP III 1999, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q6

The key idea

The equation says that $(x^{1/n})^2 + (y^{1/n})^2 = (a^{1/n})^2$, which is a circle in disguise. So put $x^{1/n} = a^{1/n}\cos t$ and $y^{1/n} = a^{1/n}\sin t$, exactly as we would for a circle, and then raise to the nth power. The fact that n is odd is what lets us take real nth roots of negative numbers, so the curve lives in all four quadrants.

Solution

The parametrisation

Because n is odd, every real number x has exactly one real nth root $x^{1/n}$, with the same sign as x, and $x^{2/n}$ means $(x^{1/n})^2$. Writing $X = x^{1/n}$, $Y = y^{1/n}$ and $R = a^{1/n} > 0$, the equation $(*)$ becomes

$$X^2 + Y^2 = R^2,$$

a circle of radius R, which we parametrise by $X = R\cos t$, $Y = R\sin t$. Raising to the nth power (a bijection of $\mathbb{R}$, as n is odd) gives

$$\boxed{x = a\cos^n t, \qquad y = a\sin^n t, \qquad 0 \le t \le 2\pi.}$$

Every point of the curve arises this way, since every point (X, Y) of the circle does.

Ask yourself Why is this anticlockwise, rather than just "some" traversal of the curve?

The map $(X, Y) \mapsto (X^n, Y^n)$ keeps the sign of each coordinate, so it sends each quadrant to itself. As t runs from 0 to 2π, the point $(R\cos t, R\sin t)$ goes once round the circle anticlockwise, visiting the positive x-axis, the first quadrant, the positive y-axis, the second quadrant, and so on, in that order. Its image (x, y) visits the same axes and quadrants in the same order, so it also goes once round the origin anticlockwise. (Within the first quadrant, for instance, $x = a\cos^n t$ decreases from a to 0 while $y = a\sin^n t$

increases from 0 to a.) When $n = 1$ this is just the familiar circle $x = a\cos t$, $y = a\sin t$.

Sketch for $n = 3$

Now $x = a\cos^3 t$, $y = a\sin^3 t$, and the curve is $x^{2/3} + y^{2/3} = a^{2/3}$.

Symmetry. The equation is unchanged by $x \mapsto -x$, by $y \mapsto -y$ and by swapping x and y. So the curve is symmetric in both axes and in the lines $y = \pm x$, and we need only study the first quadrant, $0 \le t \le \frac{\pi}{2}$.

Intercepts. At $t = 0$ we are at $(a, 0)$ and at $t = \frac{\pi}{2}$ at $(0, a)$. By symmetry the curve meets the axes only at $(\pm a, 0)$ and $(0, \pm a)$. On the line $y = x$ (at $t = \frac{\pi}{4}$) we are at $(\frac{a}{2\sqrt{2}}, \frac{a}{2\sqrt{2}})$, only about $0.35a$ along each axis.

Gradient. We have

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -3a\cos^2 t\sin t, \qquad \frac{\mathrm{d}y}{\mathrm{d}t} = 3a\sin^2 t\cos t,$$

so for $0 < t < \frac{\pi}{2}$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{3a\sin^2 t\cos t}{-3a\cos^2 t\sin t} = -\tan t.$$

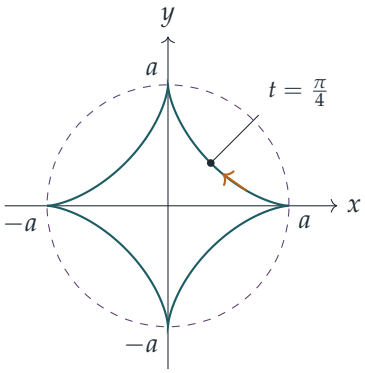
This is 0 as $t \to 0$ and tends to $-\infty$ as $t \to \frac{\pi}{2}$. So at $(a, 0)$ the curve arrives tangent to the x-axis, and at $(0, a)$ it arrives tangent to the y-axis. Both $\mathrm{d}x/\mathrm{d}t$ and $\mathrm{d}y/\mathrm{d}t$ vanish at $t = 0$ and $t = \frac{\pi}{2}$. The arc from the first quadrant meets the reflected arc from the fourth quadrant at $(a, 0)$, and both are tangent to the x-axis there, so the two arcs come together in a sharp point. The curve has *cusps* at the four points $(\pm a, 0)$, $(0, \pm a)$.

Concavity. Differentiating $-\tan t$ with respect to x,

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{-\sec^2 t}{-3a\cos^2 t\sin t} = \frac{1}{3a\cos^4 t\sin t} > 0 \quad \text{for } 0 < t < \tfrac{\pi}{2}.$$

So the first-quadrant arc is convex: it bends in towards the origin, not outwards like a circle. It lies below its chord $x + y = a$. Indeed $\cos^3 t + \sin^3 t \le \cos^2 t + \sin^2 t = 1$, with equality only at the cusps.

Putting this together gives a four-pointed star (the *astroid*). It is shown below with the circle of radius a dashed for comparison.



The area for $n = 3$

Using the derivatives above,

$$\begin{aligned} x\frac{\mathrm{d}y}{\mathrm{d}t} - y\frac{\mathrm{d}x}{\mathrm{d}t} &= a\cos^3 t \cdot 3a\sin^2 t\cos t + a\sin^3 t \cdot 3a\cos^2 t\sin t \\ &= 3a^2\sin^2 t\cos^2 t\,(\cos^2 t + \sin^2 t) \\ &= 3a^2\sin^2 t\cos^2 t = \tfrac{3}{4}a^2\sin^2 2t. \end{aligned}$$

Since $\sin^2 2t = \frac{1}{2}(1 - \cos 4t)$, and $\cos 4t$ integrates to zero over $[0, 2\pi]$,

$$\int_0^{2\pi} \sin^2 2t\,\mathrm{d}t = \pi.$$

Therefore

$$A = \frac{1}{2}\cdot\frac{3a^2}{4}\cdot\pi = \boxed{\frac{3\pi a^2}{8}}.$$

The formula gives a positive area because the curve is traversed anticlockwise, which is why the direction mattered in the first part.

Checks

- The astroid lies inside the diamond $|x| + |y| \le a$, which has area $2a^2$, so the answer must be less than $2a^2$. Indeed $\frac{3\pi}{8}a^2 \approx 1.178a^2$.
- As a check that doesn't use the formula, $4\int_0^a \left(a^{2/3} - x^{2/3}\right)^{3/2}\mathrm{d}x$ with $a = 1$, evaluated numerically, gives $1.178097\ldots$, matching $3\pi/8 = 1.1780972\ldots$.
- With $n = 1$ the same method gives $x\dot{y} - y\dot{x} = a^2$ and $A = \pi a^2$, the circle, as it should.

Common slips

- Writing $x = a\cos^n t$ without explaining why negative x is allowed. The oddness of n is exactly what makes $x^{2/n}$ meaningful (and the parametrisation complete) for $x < 0$.
- Drawing the curve bulging outwards like a squashed circle. The second derivative shows it curves inwards, with sharp cusps on the axes.
- Integrating only over $[0, \frac{\pi}{2}]$ and forgetting to multiply by 4, or making a sign error in $-y\,\mathrm{d}x/\mathrm{d}t$ (both terms end up positive).

Beyond STEP

The area formula is a special case of *Green's theorem*, which turns an integral over a region into an integral round its boundary. It is the principle behind the planimeter,

a mechanical instrument that measures area by tracing the boundary. Curves like $|x|^p + |y|^p = 1$ also give the "unit circles" of the ℓ^p norms, which you meet in analysis and linear algebra; the astroid is the case $p = \frac{2}{3}$, which is not convex, so it does not give a norm. The astroid also turns up as the path traced by a point on a circle rolling inside a circle four times its radius. For more problems in this spirit, see Siklos, *Advanced Problems in Mathematics*.

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